

PertAL: an efficient and robust active learning framework for budget-constrained single-cell perturbation screens

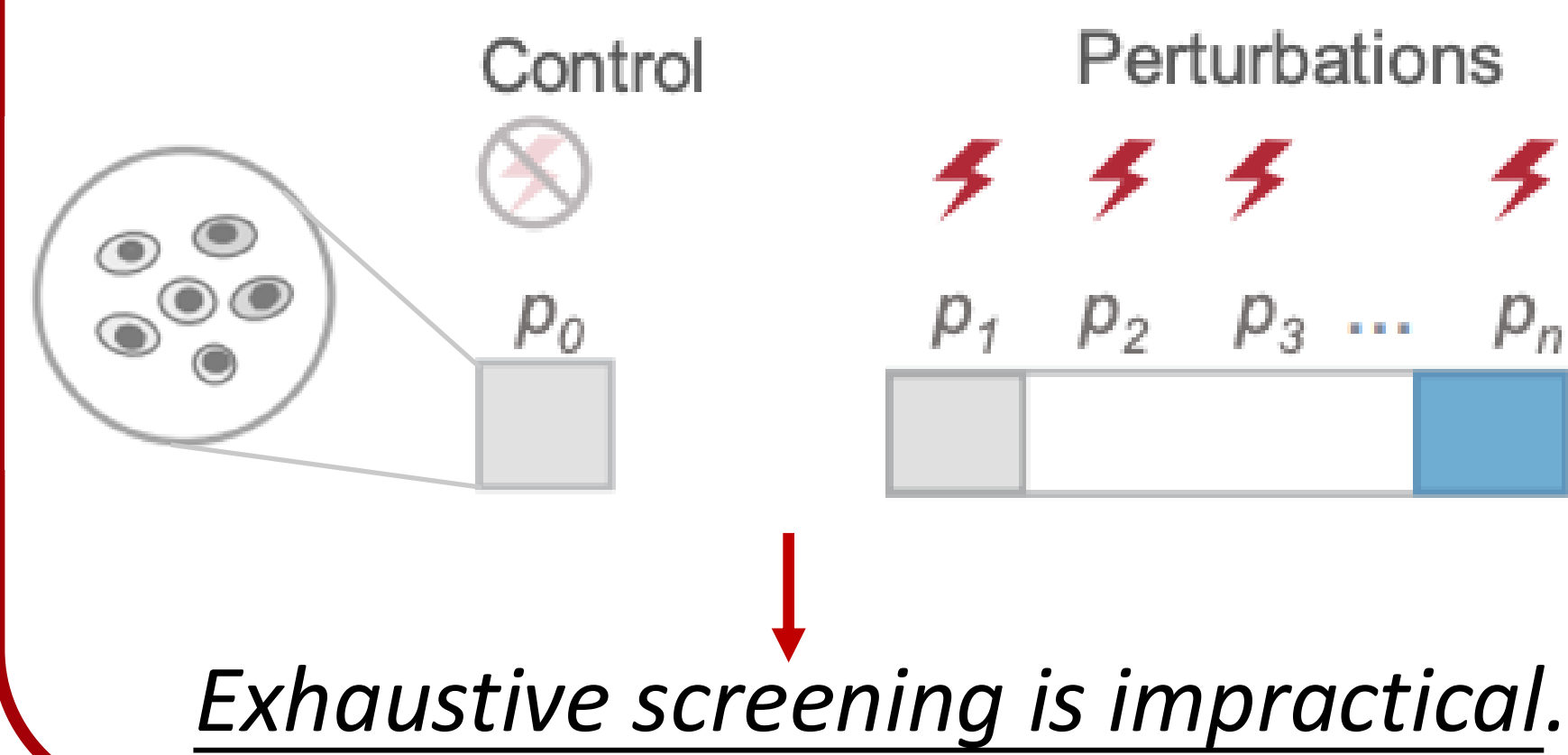
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Introduction

Single-cell perturbation screening

- Mapping cellular responses to genetic perturbations at single-cell resolution.

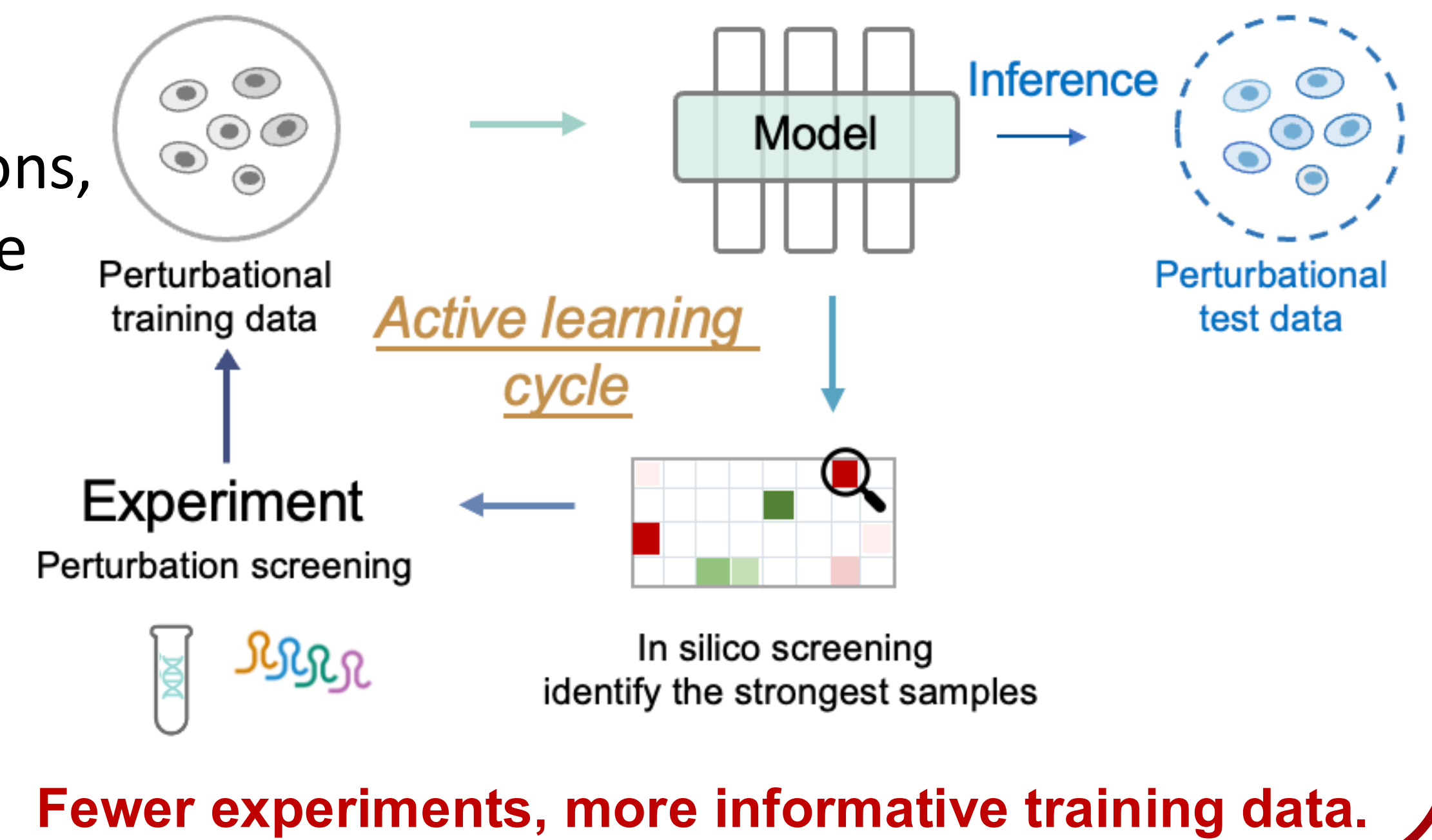


Active learning

- Iteratively select informative perturbations, measure their responses, and update the model.

Challenges

1. Limited budget
2. Technical noise mistaken for biology
3. Run-to-run biological instability

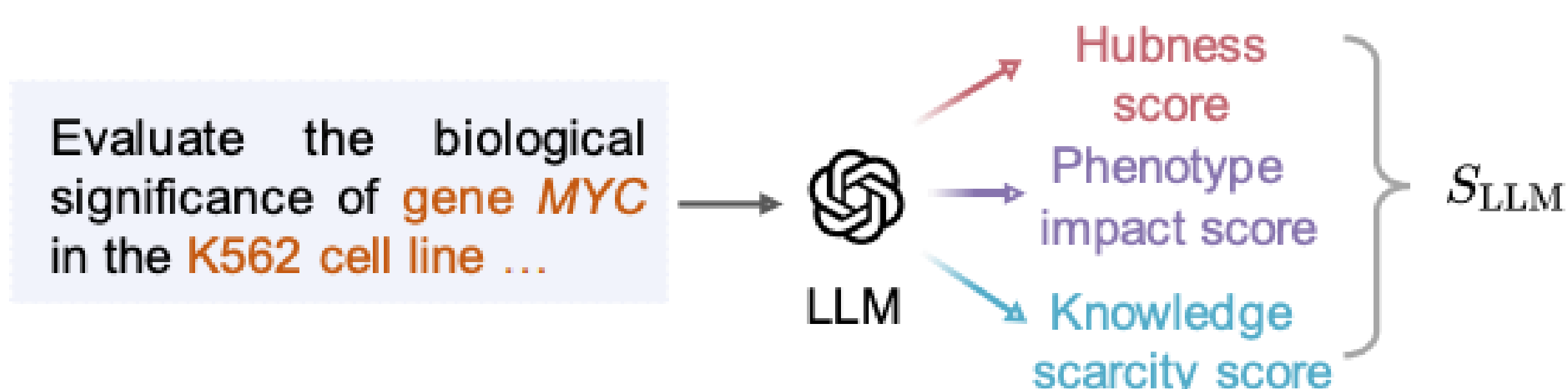


Methods

(a) LLM-driven scoring of biological reasoning

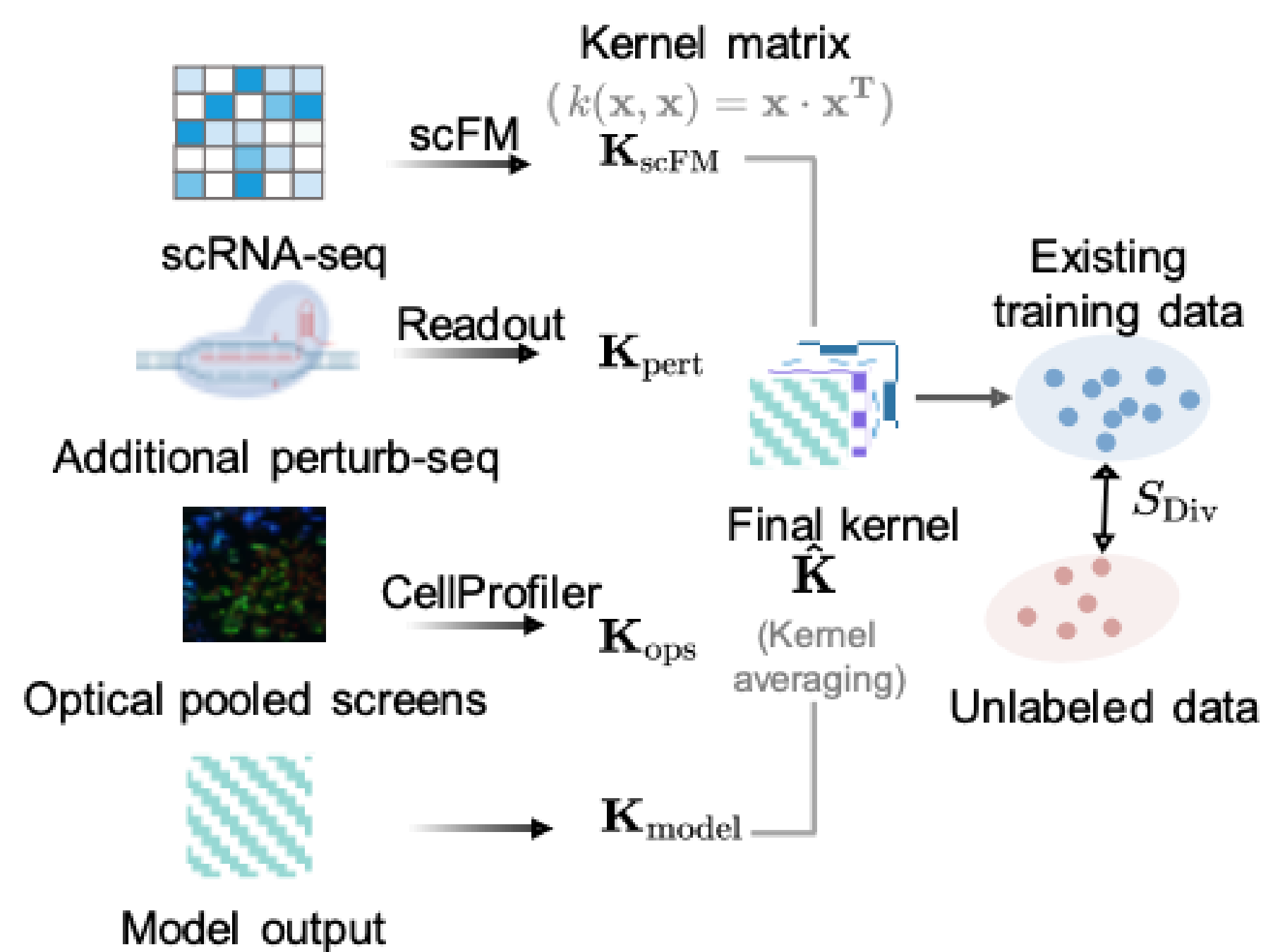
- Evaluate each candidate from three biological perspectives
- Aggregate the scores into a normalized acquisition prior

$$S_{LLM}(p_i) = \frac{S_{hub}(p_i) + S_{phe}(p_i) + S_{know}(p_i) - \mu_{total}}{\sigma_{total}}$$



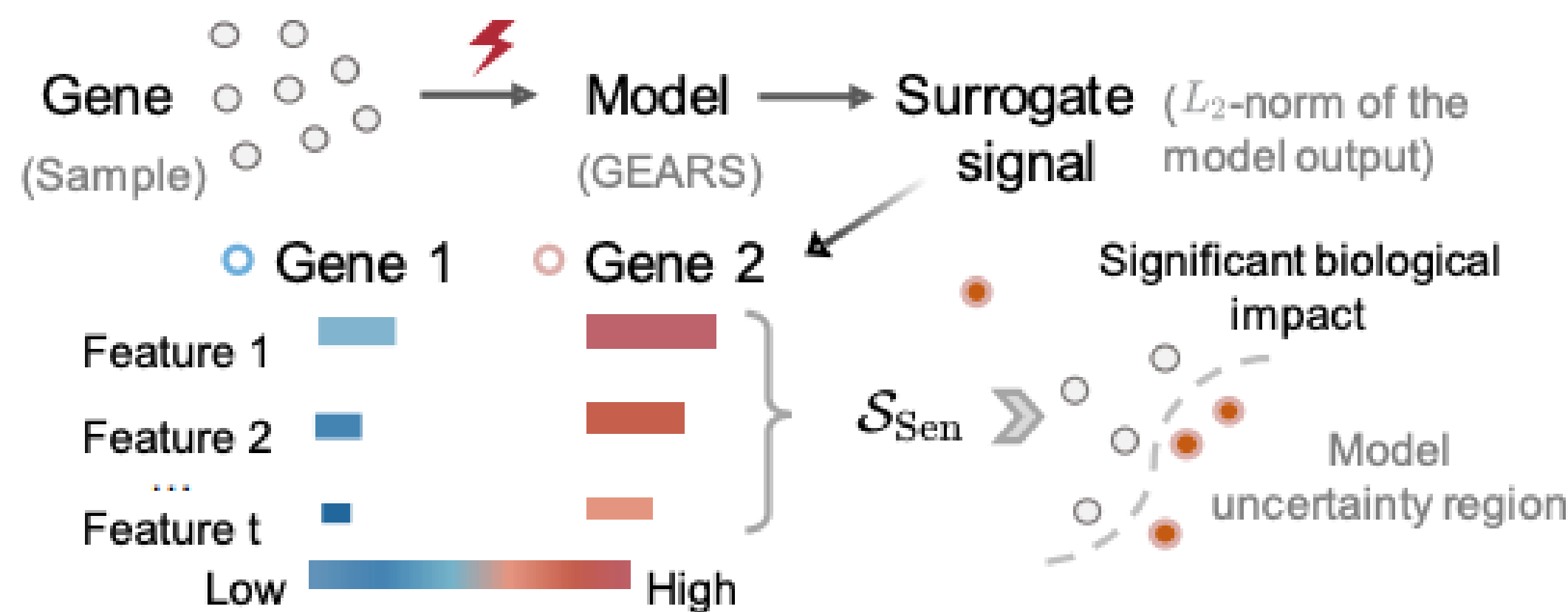
(b) Multi-view diversity assessment

- Construct and fuse similarity kernels from complementary biological and model-derived views



(c) Gradient-based sensitivity quantification

- Measure response magnitude and gradient sensitivity
- Prioritize impactful perturbations poorly captured by the model



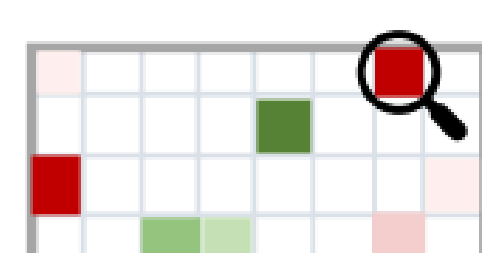
(d) Weighted scoring and selection rule

- Aggregate the three scores by weighted summation

$$S_{final}(p_i) = \alpha \cdot S_{LLM}(p_i) + \beta \cdot S_{Div}(p_i) + \gamma \cdot S_{Sen}(p_i)$$

- Greedily select the highest-scoring perturbations

$$p^* = \arg \max_{i \in D_{unlabeled}} S_{final}(p_i)$$



Experimental Results

Experimental setup

Perturbation-level splits

Dataset	No. of cells	No. of perturbation conditions	No. of control cells	No. of perturbed cells
Replogle K562	302,876	1,998	10,691	292,185
Replogle RPE1	131,201	1,280	11,485	119,716
Adamson	61,939	72	24,263	37,676

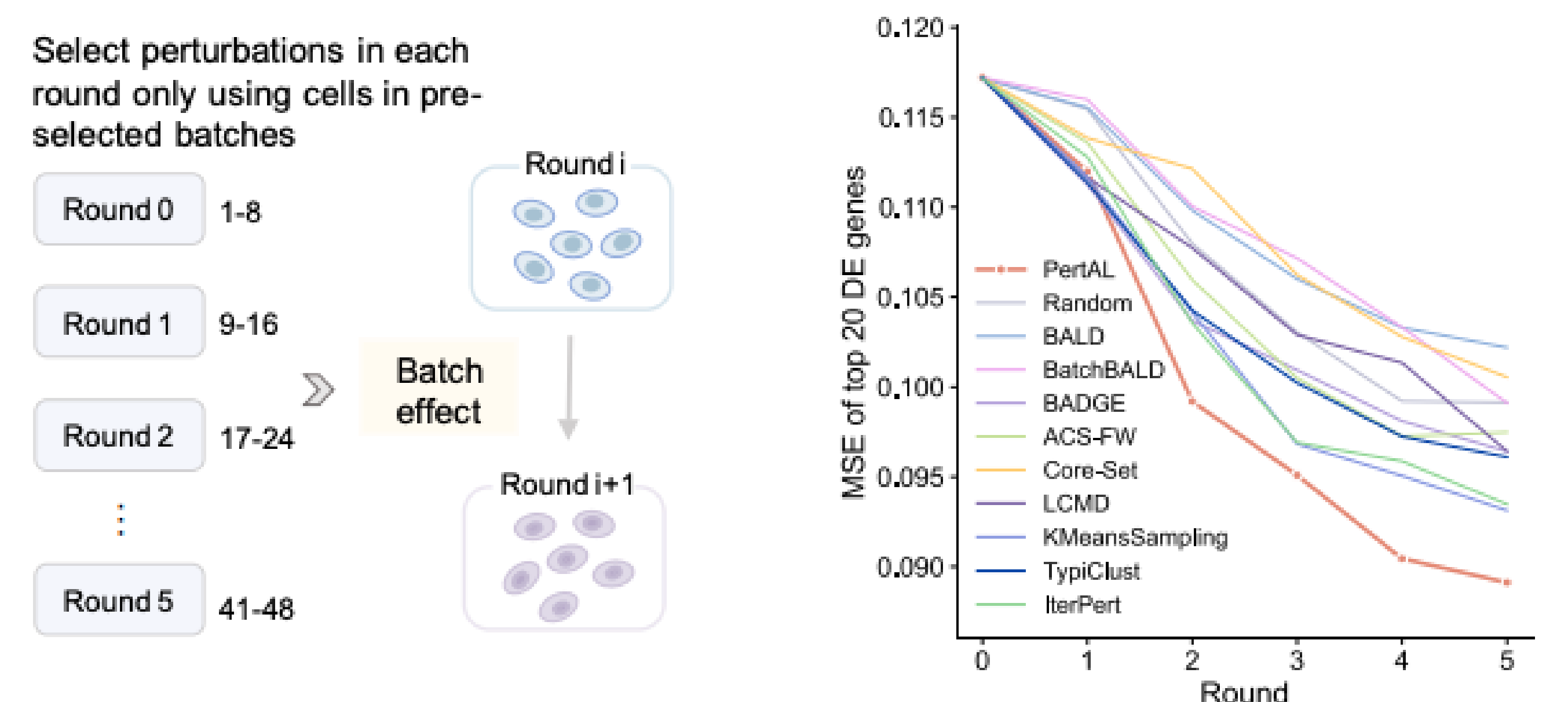
- 5 active learning rounds;
- Predictive model: GEARS
- Budget (# perturb.) per round: Replogle K562/RPE1 (100); Adamson (5)

Results

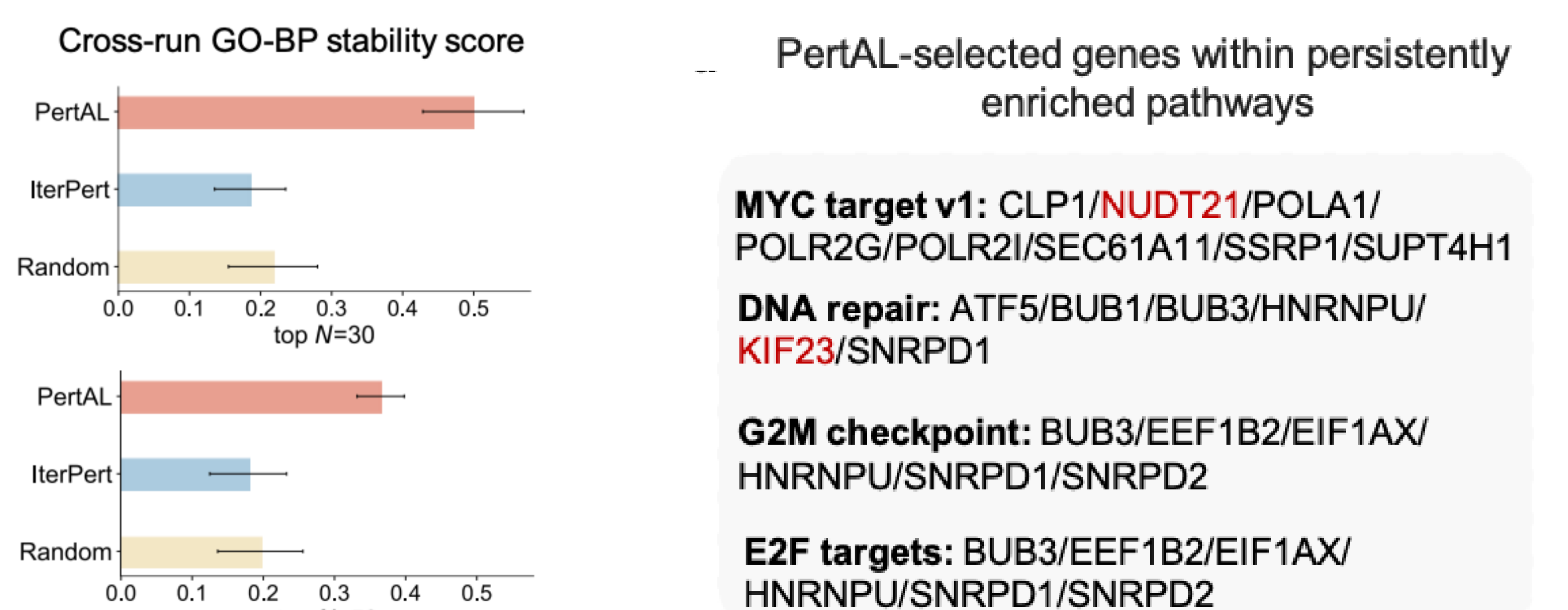
- PertAL outperforms 10 state-of-the-art AL baselines

Model	Replogle K562		Replogle RPE1		Adamson	
	Pearson ($\uparrow$)	MSE $\times 10^2$ ($\downarrow$)	Pearson ($\uparrow$)	MSE $\times 10^2$ ($\downarrow$)	Pearson ($\uparrow$)	MSE $\times 10^2$ ($\downarrow$)
Random	0.362 $\pm$ 0.022	8.840 $\pm$ 1.330	0.778 $\pm$ 0.021	3.300 $\pm$ 0.390	0.825 $\pm$ 0.094	1.820 $\pm$ 1.190
BALD [6]	0.318 $\pm$ 0.032	9.110 $\pm$ 1.560	0.776 $\pm$ 0.014	3.290 $\pm$ 0.390	0.830 $\pm$ 0.068	1.800 $\pm$ 1.000
BatchBALD [16]	0.341 $\pm$ 0.016	9.000 $\pm$ 1.460	0.785 $\pm$ 0.014	3.400 $\pm$ 0.440	0.814 $\pm$ 0.110	1.910 $\pm$ 1.140
BADGE [17]	0.359 $\pm$ 0.030	8.470 $\pm$ 0.910	0.780 $\pm$ 0.020	3.410 $\pm$ 0.330	0.792 $\pm$ 0.131	1.920 $\pm$ 1.110
ACS-FW [18]	0.362 $\pm$ 0.021	8.260 $\pm$ 1.350	0.775 $\pm$ 0.019	3.350 $\pm$ 0.420	0.760 $\pm$ 0.181	4.640 $\pm$ 5.760
Core-Set [19]	0.321 $\pm$ 0.016	9.260 $\pm$ 1.810	0.771 $\pm$ 0.015	3.370 $\pm$ 0.240	0.822 $\pm$ 0.075	1.930 $\pm$ 1.060
LCMD [14]	0.348 $\pm$ 0.024	8.690 $\pm$ 1.250	0.781 $\pm$ 0.010	3.310 $\pm$ 0.390	0.823 $\pm$ 0.082	1.920 $\pm$ 1.160
KMeansSampling [20]	0.359 $\pm$ 0.052	8.390 $\pm$ 1.390	0.781 $\pm$ 0.023	3.330 $\pm$ 0.410	0.782 $\pm$ 0.130	2.060 $\pm$ 1.380
TypiClust [21]	0.364 $\pm$ 0.029	8.150 $\pm$ 1.320	0.779 $\pm$ 0.021	3.280 $\pm$ 0.320	0.722 $\pm$ 0.053	2.190 $\pm$ 0.950
IterPert [7]	0.362 $\pm$ 0.021	8.040 $\pm$ 1.090	0.781 $\pm$ 0.021	3.410 $\pm$ 0.480	0.826 $\pm$ 0.092	1.920 $\pm$ 1.160
PertAL (Ours)	0.392 $\pm$ 0.021	7.900 $\pm$ 1.150	0.789 $\pm$ 0.017	3.240 $\pm$ 0.470	0.853 $\pm$ 0.081	1.640 $\pm$ 1.110

- PertAL obtains robust performance against batch effects



- PertAL consistently converges on stable and biologically meaningful genes across runs



Conclusion

- We propose PertAL, a label-efficient active learning framework for budget-constrained single-cell perturbation screens.
- It outperforms 10 active learning baselines across various scenarios.